Do Now · retrieval

Lesson 136 · Normalisation: First and Second Normal Form

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What table resolves a many-to-many database relationship?

Q2

What is a composite key?

Q3

When does a linear search stop?

Q4

In Big O notation, what is binary search worst-case complexity?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Make fields atomic — Replace 'Maths, Art' in one cell with separate enrolment records.

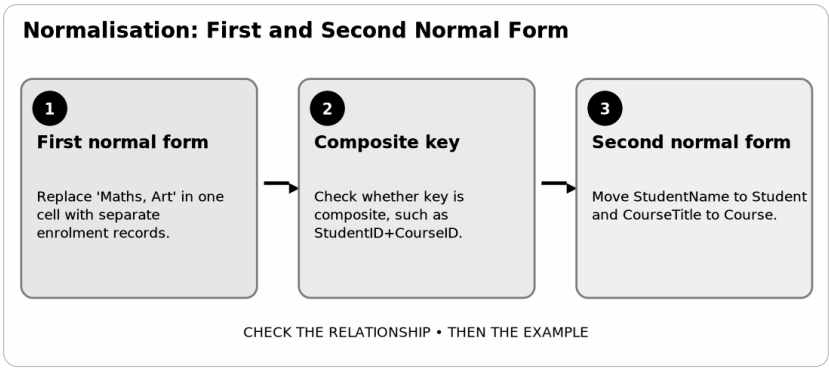
Core idea 2

Identify keys — Check whether key is composite, such as StudentID+CourseID.

Core idea 3

Remove partial dependencies — Move StudentName to Student and CourseTitle to Course.

Key visual · complete concept



New Learning · applying the ideas

Vocabulary · use the correct technical terms

1NF — Attributes hold atomic values and rows can be uniquely identified.

2NF — In 1NF, no non-key attribute depends on only part of a composite candidate key.

partial dependency — Non-key field depends on a subset of a composite key.

anomaly — Insertion, update or deletion causes undesirable effects.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Enrolment(StudentID,CourseID,StudentName,CourseName,Grade) has composite PK (StudentID,CourseID). Achieve 2NF.

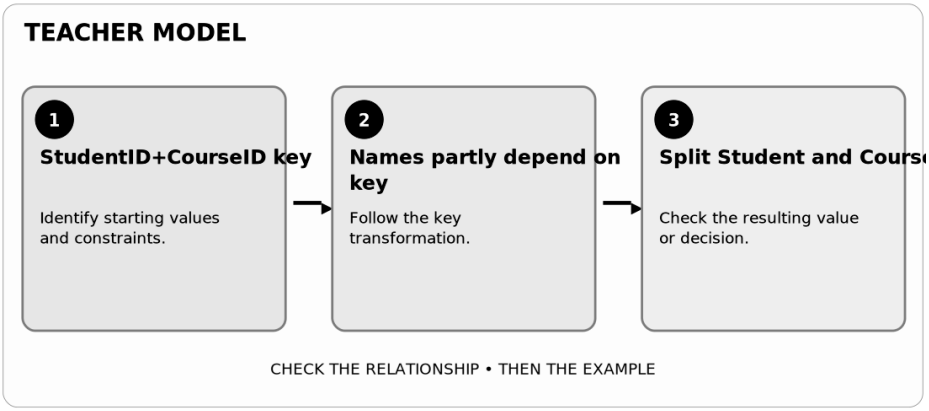
Follow the worked method

1. StudentID+CourseID key — Find the given data and name the required result.

2. Names partly depend on key — Explain why this intermediate stage is necessary.

3. Split Student and Course — StudentName depends only on StudentID; CourseName only on CourseID. Split into Student(StudentID PK,StudentName), Course(CourseID PK,CourseName), Enrolment(StudentID FK,CourseID FK,Grade) with composite PK.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Composite OrderID+ProductID
Find the given data and name the required result.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain atomicity in 1NF and correct a cell containing 'Red, Blue, Green'. [6]

Question 2

Explain partial dependency using OrderID+ProductID key and ProductDescription. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Convert StudentID,ClubID,StudentName,ClubName,JoinDate with composite key to 2NF. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A relation with a single-attribute candidate key can have a partial dependency on that key.

Use key vocabulary from memory

1. Explain 1NF in your own words.

2. Explain 2NF in your own words.

3. Explain partial dependency in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.